Self-Supervised Hyperspectral Anomaly Detection Based on Finite Spatialwise Attention

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***Abstract*— Hyperspectral anomaly detection (HAD) is of great value in both practical and theoretical terms. However, due to the lack of available semantic labels, previous works mainly relied on unsupervised or semisupervised methods to construct learning models, which inevitably lacked semantic guidance and led to limited AD effectiveness. Besides, few previous methods jointly mine spectral and spatial global dependencies, which limits their effectiveness in practical scenarios. To address the above-mentioned problems, we design a novel self-supervised HAD method, named the Self-Supervised HAD method based on the finite spatialwise attention (FSA). The core of the proposed method is the designed self-supervised HAD transFormer (SSHADFormer). It explores the specific spectral attributes of hyperspectral images (HSIs) to reconstruct background HSI from a given RGB image, which solves the difficulty of acquiring semantic information and enhances the agility of AD models. In addition, we propose an FSA mechanism. The mechanism mines the cluster structure of the background spectrum in a data-driven manner, enhancing the discriminative ability between background and anomalous targets while avoiding anomalous target interference during training. Extensive experiments on six public datasets demonstrate the effectiveness and agility of the proposed method.**

***Index Terms*— Hyperspectral anomaly detection (HAD), self-attention mechanism, self-supervised learning, spatial cluster, transformer.**

I. Introduction

HYPERSPECTRAL imagery is crucial for remote sensing missions and has a wide range of practical applications, including environmental monitoring and resource surveys. Hyperspectral anomaly detection (HAD) is a highly challenging task that aims to identify anomalous target positions that differ spectrally and cosmetically from the background pixels. The lack of predefined anomaly labels during the training stage presents significant challenges to algorithm design and makes HAD a valuable research topic.

In recent years, there has been a surge in well-designed HAD methods. These methods can be broadly categorized into three groups: background model-based (model-based), representation-based (representation-based), and deep learning-based methods. The Reed–Xiaoli (RX) method [1] is regarded as the legend in model-based methods, which models the background spectral into a multivariate Gaussian distribution and detects anomalies through Mahalanobis distance. Several improved RX methods have since been proposed, including the kernel versions of RX (KRX) [2], the local-RX (LRX) [3], the weighted-RX (WRX) [4], and the local summation AD method (LSAD) [5], all of which have improved the accuracy of model-based methods. Additionally, statistical enhancement model-based methods have been designed, assuming a low probability of anomalies occurring in hyperspectral image (HSI) [6], [7], [8]. Furthermore, some robust model-based methods have been proposed that integrate robust statistical strategies to improve the robustness of the background model [9]. The kernel-based learning and robust regression analysis can be well integrated, and the robust nonlinear AD (RNAD) method [10] has developed, including a nonlinear robust detection strategy.

In addition to the model-based approach, representation-based methods aim to design a more efficient feature representation from the perspectives of collaborative representation [11], entropy-based representation [12], sparse representation [13], [14], [15], [16], [17], [18], and some feature learning methods, such as feature transformation [19] and eigendecomposition [20]. For instance, the collaborative representation for anomaly detector (CRD) [11] regards the pixel as normal if it can be represented by the adjacent surrounding neighborhoods. Some sparse coding methods have also achieved considerable performance by integrating background prior knowledge into feature learning methods and identifying anomalous targets by feature-level distance. The low-rank and sparse representation (LRASR) method [21] proposed an effective sparse coding-based AD that explores the low-rank property of the background. The sparsity score estimation framework for HAD (SSEAD) [22] is designed, which emphasizes the importance of dictionary atom frequency for background reconstruction.

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More recently, deep learning-driven HAD methods have rapidly developed. A series of deep learning models suitable for HAD have been proposed. For instance, Jiang et al. [23] proposed a semisupervised GAN-based model. By reconstructing the spectral information of the background seed points, the semisupervised model can obtain the distribution of the background spectrum. Liu et al. [24] presented a dual-frequency autoencoder (DFAE) that decomposed the task into two parallel reconstruction tasks and detected anomalous targets in the images by fusing high-frequency and low-frequency reconstruction results. Besides the above methods, more generative models [25], [26], [27] have been proposed for HAD tasks, as illustrated in detail in Section II-A.

Due to the absence of semantic labels, it is not feasible to establish an effective supervised learning method for HAD. Consequently, the majority of previous HAD methods can be categorized as unsupervised or semisupervised approaches. During the training phase, the lack of guiding semantic information hampers the model's discriminative ability. Recently, self-supervised methods have been proposed for image AD that mines the inherent object attributes in natural scenes to construct pretext tasks suitable for AD. For instance, Li et al. [28] mined the structural distribution pattern of normal objects through content erasure and paste operations and regarded samples that broke the pattern as anomalous. In addition, an existing study [29] has demonstrated that self-supervised learning is a viable way to capture high-level semantic information and achieve optimal results on image AD tasks. Therefore, constructing a mapping between attributes and content is critical to developing a self-supervised AD method. An effective self-supervised mapping will establish an appropriate supervised relationship between the normal object and attribute. Importantly, in addition to the structural and posture attributes frequently employed in natural images, HSI also contains a spectral attribute that is a unique advantage for developing a self-supervised method. However, in the field of HAD, a self-supervised HAD method has not yet been proposed, and the specific pipeline remains to be further explored.

There is a close relationship between spectral and spatial dimensions in HSIs, indicating that background features should exhibit long-range dependency and possess low-rank properties [20]. However, most existing deep HAD methods treat pixelwise spectral reconstruction as the training objective [23], [30], resulting in a small spatial receptive field that fails to capture the intrinsic relationship between spectral and spatial dimensions. Moreover, pixelwise spectral reconstruction techniques are unable to learn the spatial distribution properties of the background, relying solely on classic background extraction algorithms for the utilization of spatial information. Consequently, most current HAD methods are incapable of jointly mining spectral and spatial information in a data-driven manner, thereby leading to limited applicability due to the background extraction prior. Therefore, there is a need to investigate how to aggregate background features by jointly exploiting spectral and spatial information while also taking advantage of the long-range dependency of background pixels.

To address the aforementioned issues, we propose a self-supervised HAD method based on finite spatialwise attention (FSA) in this study. The motivation for our work is illustrated in Fig. 1. Specifically, the accuracy of pixel-level spectral reconstruction methods is limited by the spectral reconstruction error, especially when the distance between the abnormal spectrum (red curve) and the normal spectrum (green curve) is much smaller than the distance between normal pixel points [green curve with other normal curves in Fig. 1(b)]. In such cases, the decision boundary may be unclear and lead to misjudgment if errors occur in the reconstruction process. Furthermore, as observed in Fig. 1(b), there is a clear difference in appearance, color, and spectrum between the red and green points, which are intrinsic attributes of objects. However, pixel-level spectral reconstruction methods only utilize a single-spectral property and shallow information of HSI, as shown in Fig. 1(a). In contrast, our proposed method establishes relationships between attributes of normal pixels to obtain high-level semantic information, as depicted in Fig. 1(c).

To extract semantic information from HSI without anomaly labels, we propose the self-supervised HAD transformer (SSHADFormer). This method capitalizes on the attention mechanism and integrates the statistical distribution of the background, thereby forming a mapping process for semantic attributes specific to the HAD task. By reconstructing the background HSI from an RGB image, SSHADFormer establishes the relationship between the appearance and spectral attributes of normal pixels, allowing it to identify abnormal pixels that break this relationship. Through self-supervised learning, SSHADFormer acquires high-level semantics that enhance the agility and discriminative power of the AD model. Furthermore, we introduce the FSA mechanism. First, FSA builds global spatial dependencies and explores the clustering structure of the background spectrum. Second, taking advantage of the low-rank property of the background and the sparsity of anomalous objects, FSA distinguishes background pixels from anomalous ones. Pixels that are close to the cluster center are likely to belong to the background, while those far from the center with high local attention energy may be an anomalous target. This approach avoids the interference of anomalous targets during the training stage and enhances the discrimination between background and anomaly. Finally, during the testing phase, SSHADFormer detects abnormal targets based on recovery error and foreground score.

The main contributions of this work are summarized as follows.

1) We introduce self-supervised learning into HAD, enabling the model to acquire semantic information and enhance discrimination by bridging appearance and spectral attributes.
2) We propose a novel transformer architecture named SSHADFormer, which harnesses the global feature extraction capabilities of the transformer network guided by both global spatial and spectral information in an end-to-end fashion.
3) We design the FSA mechanism to mine the cluster structure of the background spectrum in a data-driven

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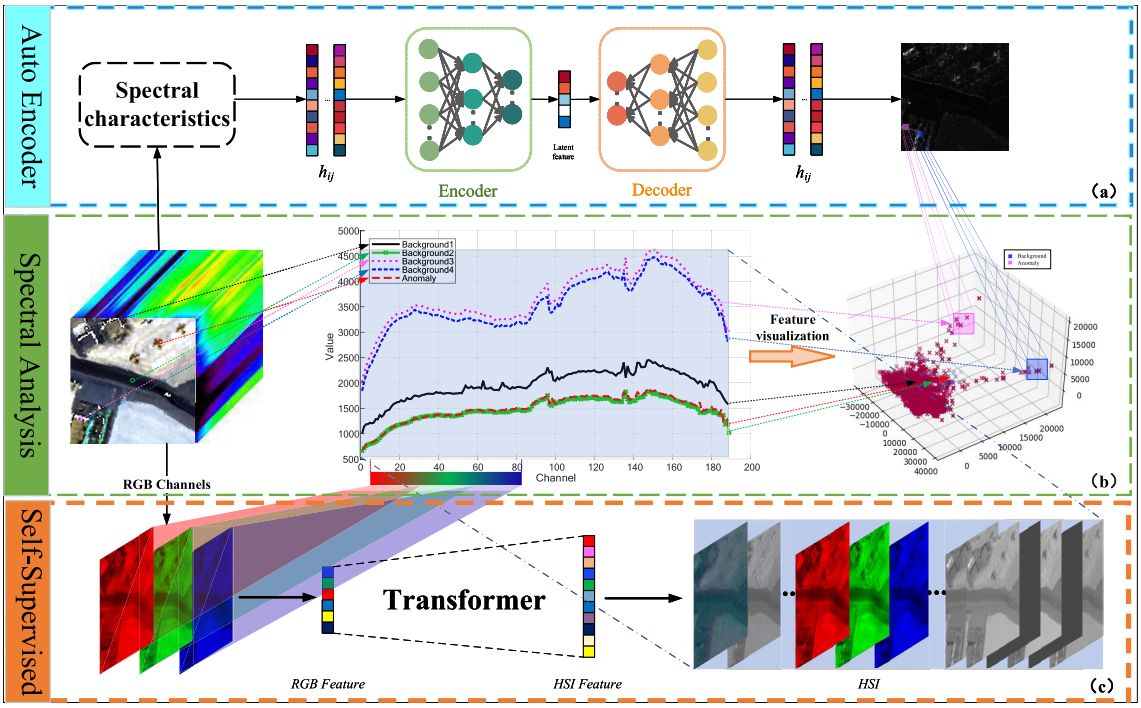


Fig. 1. Motivation of the proposed method. (a) Detection process of the pixel-level spectral reconstruction method. (b) Real-spectral curves of multiple positions on the image and the visual distribution of spectral distance between pixels. (c) Detection schematic of the proposed method.

manner. The FSA mechanism allows the model to be free from anomalous target interference during training and increases the distinction between the background and anomalous targets.

We validate the proposed method on six public datasets, demonstrating its advanced and generalizability. Our method also has good agility and discrimination.

The remainder of this article is structured as follows. Section II provides a succinct overview of related works. In Section III, the proposed self-supervised AD framework is elaborated. Section IV presents the conducted experiments and their analysis. Ultimately, the conclusions drawn from our study are encapsulated in Section V.

II. Related Works

A. Deep Learning-Based Hyperspectral AD

In this section, we present an overview of deep learning-based methods for HAD. Deep learning methods have gained significant attention due to their powerful data representation capabilities. Li et al. [31] proposed a transferred deep convolutional neural network (CNN) for AD. Song et al. [32] designed a CNN model to obtain abundance maps, while Fu et al. [33] combined a CNN denoiser with a background dictionary construction method to improve detection results. Recently, generative spectral reconstruction-based methods have become the mainstream direction. Jiang et al. [30] proposed an unsupervised discriminative reconstruction constrained generative adversarial network (HADGAN) for HAD. Jiang et al. [23] presented a novel semisupervised spectral learning framework based on GAN, which uses knowledge of the spectral distribution to build a learning estimation module for AD. Liu et al. [24] introduced a DFAE model that transforms the original HSI into high- and low-frequency components before detection. However, most of these methods are unsupervised and rely on prior assumptions, making them less reliable. On the other hand, self-supervised learning provides a data-driven solution for the under-labeled problem. Unfortunately, self-supervised learning has not been explored in the field of HAD. Therefore, our proposed method introduces self-supervised learning to HAD for the first time and achieves better performance.

B. Transformer

A transformer is a type of deep neural network that relies on the self-attention mechanism. It was initially applied to natural language processing (NLP) tasks, leading to significant improvements [34], [35], [36]. Motivated by its powerful representation capabilities, researchers are exploring ways to apply the transformer to computer vision tasks, such as object detection [37] and low-level visual tasks [38]. The visual transformer (ViT) model proposed by Dosovitskiy et al. [39], which directly applies a pure transformer to sequences of image patches, achieved state-of-the-art performance on multiple image recognition benchmarks. In the field of image AD, Pirnay and Chai [40] have recently proposed a self-attention approach based on pure transformers to solve AD as a patch-inpainting problem, discarding convolutions. In recent years, the transformer has been increasingly used in the field of HSI processing [41], [42], [43], [44], [45], yielding promising results. For example, in HSI classification, Hong et al. [41] proposed a transformer-based approach that reconsiders HS image classification from a sequential perspective, with modules that learn spectrally local sequence information and convey memory-like components from shallow to deep layers, respectively. In hyperspectral target detection, Rao et al. [46] proposed a Siamese transformer

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network that effectively extracts global features from spectra with long-range dependencies to achieve target detection. In HSI denoising, Wang et al. [47] developed a novel deep learning model for HSI stripe noise removal based on a semisupervised training strategy. Due to the outstanding performance of the transformer, more and more transformer-based models have been proposed to enhance various vision tasks. However, the transformer architecture has not yet been extensively studied and analyzed for HAD.

III. Methodology

A. Overall Architecture

The overall architecture of the proposed method is illustrated in Fig. 2. As depicted, our approach comprises three crucial components, namely: color cube generation, the SSHADFormer network, and construction of self-supervised learning objective.

1) Color Cube Generation: Assuming $X \in \mathbb{R}^{W\times H\times 3}$ represents the RGB image associated with the HSI from the given dataset, where W and H denote the pixel count in the width and height directions of the color image, respectively. We utilize RGB cubes as the input in this algorithm. The objective of the color cube generation operation is to construct a color cube suitable for training the network.

To satisfy the training sample requirements for the transformer network, a sliding window data augmentation technique is employed during the training phase. More specifically, the RGB image X is segmented into a sequence of overlapping color cubes $X_i \in \mathbb{R}^{k\times k\times 3}$ using a sliding window, where k indicates the block size of the color cube, $i \in, \ldots, n$ denotes the cube index number, n represents the number of color cubes, and the block size is required to satisfy $k < W$ and $k < H$. The following equation details the sliding window process:

$$\begin{cases} X_i = X[w_i + k, h_i + k, :] \\ \exists w_{i-1} + p + k < W : w_i = w_{i-1} + p, h_i = h_{i-1} \\ \exists w_{i-1} + p + k \geq W : w_i = 1, h_i = h_{i-1} + p \end{cases} \tag{1}$$

where w_i and h_i denote the starting position index of the ith color cube, respectively, and p signifies the sliding window step size. During implementation, we conduct the horizontal sliding operation initially using step size s, followed by the vertical sliding operation using the same step size s. The generated color cubes serve as the basis for the network to explore spatial and spectral information.

2) SSHADFormer: Given a training RGB sample X_i, the SSHADFormer aims to recover the background spectrum and generate a high-quality background HSI cube. Since the training samples exist in the RGB space, their content captures the appearance and structural details of the scene. If the network successfully learns the mapping from RGB to HSI for normal targets, it establishes an intrinsic connection between the appearance and spectral spaces of normal targets. It is noteworthy that both appearance structure and spectral property are inherent attributes of an object. The constructed self-supervised learning task forces the network model to capture the association between these inherent attributes. The inclusion of more normal object attribute information during training leads to better capture of high-level semantic information of normal targets [48]. Consequently, the model becomes adept at describing normal targets, leading to different recovery effects for normal and abnormal targets. Thus, it demonstrates strong discriminative and agility for HAD tasks. However, obtaining pure normal images from HAD datasets is incredibly difficult. Eliminating the interference of abnormal targets is also a critical factor in model design. To address this challenge, the SSHADFormer incorporates an FSA block (FSAB), which avoids anomalous target interference during the training stage.

The schematic of the SSHADFormer network is given in Fig. 2. The network employs a stacked transformer architecture, similar to the multistage spatialwise transformer (MST) [49]. Specifically, SSHADFormer contains N_t identical subtransformer. For simplicity, subtransformers with the same modular structure are collectively referred to as SSHADFormer. Moreover, each SSHADFormer adopts an AE structure, comprising an encoder and a decoder. The encoder and decoder both employ FSABs and skip connections are added between certain layers in the encoder and corresponding decoder layers to improve the quality of spectral recovery, following the settings in [48] and [49].

To begin, the network expands the channel count of a given training sample $X_i \in \mathbb{R}^{k\times k\times 3}$ to n_s via a convolution operation, generating $X_i^0 \in \mathbb{R}^{k\times k\times n_s}$. The resulting feature map X_i^0 is then inputted into five consecutive FSABs, each consisting of a cascaded four-layer structure comprising a convolutional/deconvolutional layer (Conv/Deconv), an FSA layer, a spectral attentive layer (SA), and a feedforward network (FF), as illustrated in Fig. 2. Layer normalization (LN) is also incorporated in the FSAB to enhance the model's overall robustness. The convolutional layer in FSAB1 is primarily employed for feature learning and smoothing, while the convolutional layers in the FSAB2 and FSAB3 modules are mainly used for spatial downsampling and generate the corresponding feature maps $X_i^2 \in \mathbb{R}^{(k/2)\times(k/2)\times 2\ n_s}$ and $X_i^3 \in \mathbb{R}^{(k_0/4)\times(k_0/4)\times 4\ n_s}$. The deconvolutional layers in FSAB4 and FSAB5 are employed for upsampling, ultimately producing the HSI $\widehat{S}_i \in \mathbb{R}^{k\times k\times n_s}$ after twice up-dimensional recovery. Throughout this process, SSHADFormer learns to extract both spatial and spectral information from the training sample, allowing for end-to-end training.

3) Construction of Self-Supervised Learning Objective: Construction of Self-supervised learning objective aims to generate a real HSI cube, denoted as S_i, which serves as the training target for the model. However, HSIs typically have a large spectral dimension, which not only incurs significant computational cost but also diminishes the model's ability to capture inter-channel dependencies. This undermines the stability of spectralwise attention and hinders the effectiveness of self-supervised learning.

In our proposed method, we address the issue of large spectral dimensions in HSIs by reducing the number of redundant spectral channels in the real HSI. Specifically, we apply the optimal clustering framework (OCF) [50] method to the

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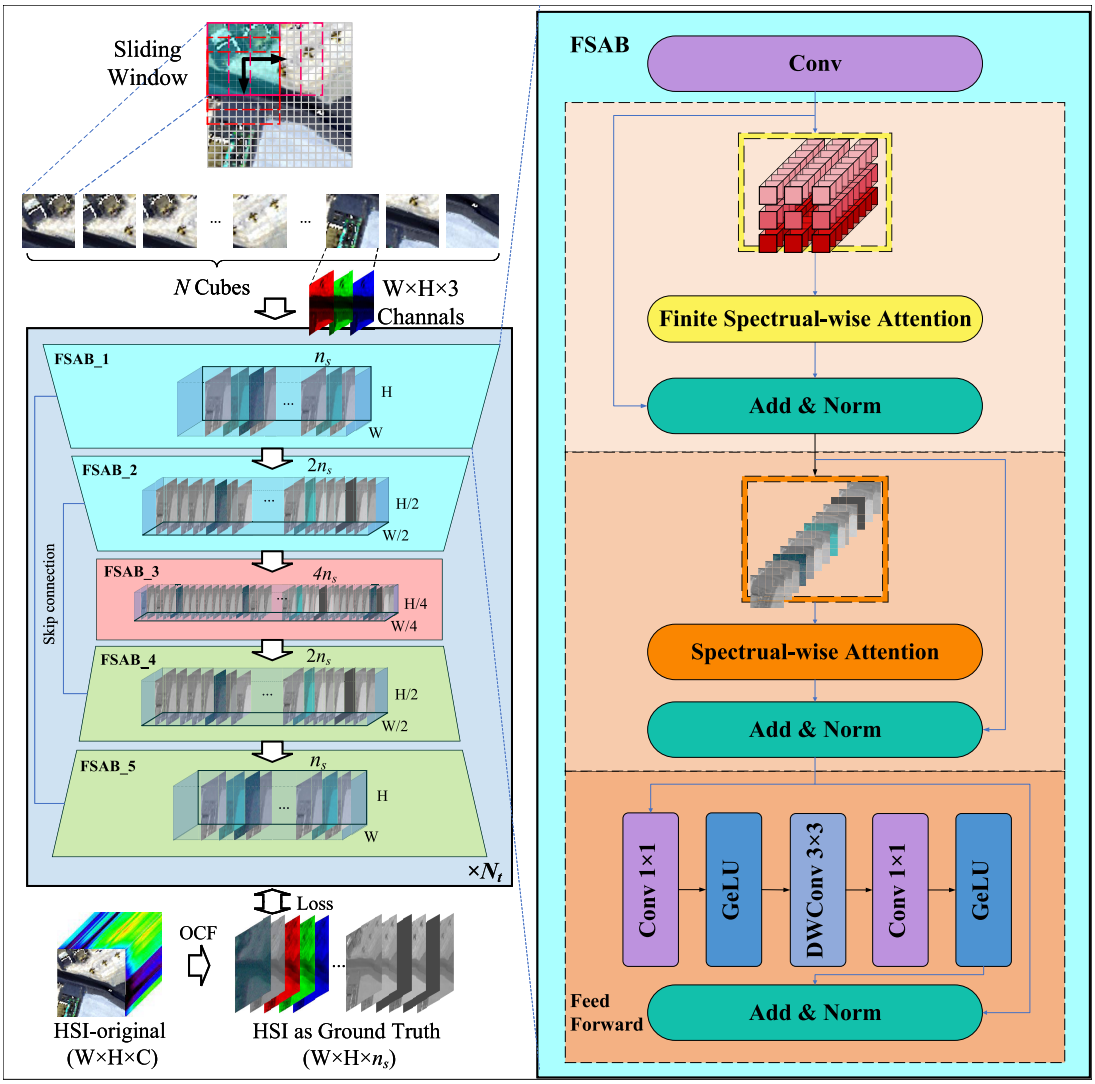


Fig. 2. Schematic of the proposed SSHADFormer. Left—A given sample is segmented into RGB cubes by a sliding window operation. Afterward, these cubes are fed into SSHADFormer. Finally, the HSIs of the background, selected by OCF, are recovered. Right—The framework of the proposed FSAB, which is the core module of SSHADFormer.

real HSI $S \in \mathbb{R}^{W\times H\times C}$ to select the most representative n_s spectral channels, resulting in a reduced $S' \in \mathbb{R}^{W\times H\times n_s}$. Then, we use a sliding window operation to generate the tagged HSI cube $S'_i \in \mathbb{R}^{k\times k\times n_s}$, corresponding to X_i. This operation is similar to the process shown in (1), and thus, will not be repeated here. OCF is a clustering-based spectral selection approach that selects typical spectral channels based on clustering and ranking results and has been widely used in hyperspectral analysis [51], [52]. The obtained hyperspectral cubes S'_i are then used to guide the network in accomplishing self-supervised learning. This channel selection process can conserve computational resources while having as little impact on performance as possible.

B. Finite Spatialwise Attention Mechanism

In our proposed method, we introduce the FSA mechanism to enhance the model's ability to distinguish between the background and anomalous targets during training, while avoiding the influence of the latter. First, the FSA constructs global spatial dependencies and explores the cluster partition of background pixels. Second, we design a confidence background learning strategy based on the low-rank of the background and the sparsity of the anomalous pixels. This strategy generates a confidence background weight map, which assigns a larger weight to reliable background pixels, thereby avoiding the influence of anomalous targets during the training procedure.

Besides, pixels with a low likelihood of being background may be classified as anomalies. Furthermore, due to the sparsity of anomalies, their attention distribution is more concentrated, resulting in a higher local attention energy. Conversely, the attention distribution of the background is more global, resulting in lower local attention energy. Based on these two analyses, we propose a confidence foreground weight that combines conjugate background weights and local attention energy to improve the discrimination between the background and anomalies. The composition of the FSA is illustrated in Fig. 3(a), while the pipeline of the confidence background learning strategy is explained in Fig. 3(b).

1) Finite Spatialwise Attention: The FSA consists of two steps. First, it establishes global dependencies between

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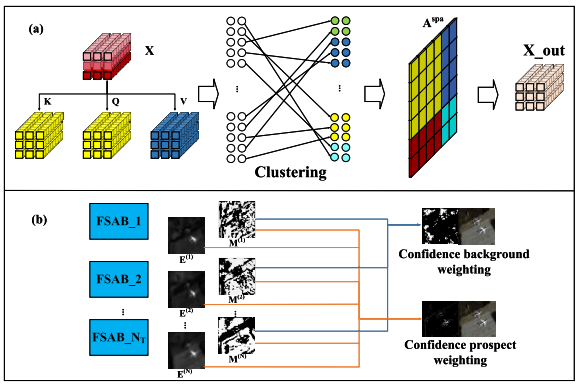


Fig. 3. Schematic of the FSA mechanism. (a) Composition structure of the FSA. (b) Diagram of confidence background learning strategy.

different location features in the spatial domain. Second, it generates corresponding spatial aggregation features based on these global dependencies.

Specifically, the given feature matrix $X \in \mathbb{R}^{k\times k\times n_s}$ is first reshaped into m tokens $X = [x_1, x_2, \ldots,]' \in \mathbb{R}^{m\times n_s}$ based on spatial coordinates, where $m = k^2$ is the number of pixels. Next, tokens X are linearly projected into query, key, and value matrices, as shown in the following:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \tag{2}$$

where $Q, K, V \in \mathbb{R}^{m\times n_s}$ represent the query, key, and value matrices, respectively, and $W_Q, W_K, W_V \in \mathbb{R}^{n_s\times n_s}$ are optimizable parameters of linear transformation. In implementation, the weights of Q and K are set to be shared, with $W_Q = W_K$.

Next, the cluster partition of the background is explored based on the low-rank of the background, after obtaining Q, K, and V. To be more precise, the matrix Q is initially clustered to obtain n_c clusters and their corresponding cluster center vectors $C = [c_1, c_2, \ldots,] \in \mathbb{R}^{n_c\times n_s}$. The clustering centers can be continuously optimized during training and can be considered as the most representative feature of the current scene when the training goal is achieved. As the background pixels are the main component in the HSI, the clustering center also represents the intrinsic background content. Then, the augmentation matrices $Q', K' = [Q; C] \in \mathbb{R}^{(m+n_c)\times n_s}$ are obtained by row concatenation. Finally, attention scores A^{spa} are calculated using the scaled dot-product attention mechanism [34]

$$A^{\text{spa}} = \text{softmax}\left(Q'K'^{\top}/\sqrt{n_s}\right) \tag{3}$$

where $K'^{\top}$ represents the transpose matrix of the augmented key matrix, and $1/(n_s)^{1/2}$ is the scaling factor.

As shown in Fig. 3, matrix $A^{\text{spa}} \in \mathbb{R}^{(m+n_c)\times(m+n_c)}$ can be divided into four components. To be specific, the first component, $A^{spa_t} = A^{\text{spa}}[1:m; 1:m] \in \mathbb{R}^{m\times m}$, represents the global attention relation between query vectors and is depicted in yellow in Fig. 3(a). A larger value of $A^{\text{spa_}t}_{i2\ j}$ indicates a strong correlation between location i and location j, which requires a higher weight during feature aggregation. The second component, $A^{\text{spa}_c} = A^{\text{spa}}[1:m; m+1:m+n_c] \in \mathbb{R}^{m\times n_c}$, represents the attention between features and cluster centers and is depicted in blue in Fig. 3(a). A larger value of $A^{\text{spa_}c}_{i2\ k}$ indicates that vector q_i, which comes from position i of the image, is more likely to fall in the kth cluster.The ith row vector of matrix A^{spa_c}, denoted as $A^{\text{spa}_c}_i$, represents the possibility that vector q_i belongs to each cluster. By finding the maximum value in $A^{\text{spa}_c}_i$, m spatial query features can be partitioned into n_c different clusters. The cluster indicator, $IA^{\text{spa}_c}_i$, is a one-hot vector with the same dimension as $A^{\text{spa}_c}_i$, where an element is assigned one only if the corresponding position is the maximum value in $A^{\text{spa}_c}_i$.

During the spatial feature aggregation process, it is crucial to ensure that the learned features possess both global and discriminative properties to recover high-quality background spectra amidst the complexity and diversity of the background. Robust global features are necessary to ensure that the model is free from distortions and noise, while diverse backgrounds require the learned features to be discriminative. To achieve this, limiting the spatial attention field to the corresponding clusters is necessary, which enables the learned features to be both discriminative and global. The calculation process can be expressed using the following:

$$\begin{cases} x'_i = \sum\limits_{j\text{in}[1,m]} \langle IA^{\text{spac}}_i, I\ A^{\text{spac}}_j\rangle A^{\text{spa_}t}_{i2\ j} \cdot v_j \\ \text{FSA}(X) = LN\big(\text{FFN}\big(X'\big)\big) + X' \end{cases} \tag{4}$$

where $\text{FSA}(X)$ represents the spatial aggregation features, $\langle\cdot,\cdot\rangle$ denotes the inner product of vectors. A value of 1 for the inner product indicates that the query vectors q_i and q_j come from the same cluster, whereas a value of 0 indicates the opposite. The intermediate variable X' represents the computational result of attention and is normalized before being fed to the feedforward network (FFN). The FFN consists of depth-separable convolutional, activation, and transformation layers. Finally, the spatial aggregation features are achieved, followed by normalization and residual connection.

FSA utilizes the K-means algorithm to cluster the query vectors, potentially due to the inverse relationship between inner product and distance, which allows for the conversion of attention score and K-means.

Particularly, if q_i and q_j originate from the same cluster with the corresponding cluster center being c_k, their attention level and Euclidean distance should satisfy the relationships shown in the following:

$$A^{\text{spa_}t}_{i2\ j} \propto \langle q_i, k_j\rangle \tag{5}$$

$$2 - 2\langle q_i, k_j\rangle = \|q_i - k_j\| \le \|q_i - c_k\| + \|k_j - c_k\| \tag{6}$$

where q_i represents the ith query vector and k_j represents the jth key vector. Thus, when q_i and k_j are close and belong to the same cluster, the corresponding attention values between them should be large [53].

Finally, the following equation presents the loss function for clustering

$$\mathcal{L}_{\text{clu}} = \sum_{k=1}^{n_c} \sum_{q_j\in c^{(k)}} \|q_j - c_k\|_2 \tag{7}$$

where $c^{(i)}$ denotes the ith cluster. An iterative approach is employed to optimize the clustering centers.

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2) Confidence Background Learning Strategy: Most existing HAD methods rely on additional hyperspectral background extraction methods [23] to discern the background distribution as prior knowledge of the scene. However, we propose a confidence background learning strategy that filters reliable background pixels in a data-driven manner using the mentioned query-to-cluster attention matrix $A^{\text{spa_}c}$. The clustering centers are considered the most representative background feature, and the kth column vector $A^{\text{spa_}c}[:, k] \in [m, 1]$ of $A^{\text{spa_}c}$ represents the probability that the corresponding position is a background pixel. Thus, we select the top r percent of elements from each cluster and generate the background weight map M based on their positions. The background weight map M can be formulated, as shown in the following:

$$M(i, j) = \sum_{k=1}^{n_c} I(i, j, k) \cdot \widehat{A^{\text{spa_}c}}(i, j, k) \tag{8}$$

where $\widehat{A^{\text{spa_}c}} \in \mathbb{R}^{k\times k\times n_c}$ is a three-dimensional matrix obtained by reshaping $A^{\text{spa_}c} \in \mathbb{R}^{m\times n_c}$ according to the spatial location. The indicator I is 1 if and only if $q_{i,j} \in c^{(k)}$ and the Euclidean distance between $q_{i,j}$ and the cluster center $c^{(k)}$ belong to the top r percent of the total elements of cluster $c^{(k)}$, otherwise, $I(i, j, k) = 0$.

Additionally, we investigate the hierarchical structure of the network and extract confidence background weight maps $M^{(1)} \sim M^{(N_T)}$ from different layers. The integration of these weight maps from each FSAB in the encoders generates the final confidence background weight map M. It is well-known that shallow features contain low-level visual information, such as appearance and texture, while deep features are more focused on content semantics. Therefore, different layers possess different abilities to discriminate between anomalies and backgrounds. During training, a reliable confidence background weight map M is required for selecting the most trustworthy background pixels. Finally, we generate the confidence background weight map by taking the intersection of $M^{(1)} \sim M^{(N_T)}$, as defined in the following:

$$\begin{aligned} M_{i,j} &= M_{i,j}^{(0)} \cap M_{i,j}^{(1)} \cdots \cap M_{i,j}^{(N_T)} \\ &= \begin{cases} 0, & M_{i,j}^{(1)} \times \cdots M_{i,j}^{(N_T)} = 0 \\ \dfrac{1}{N_T}\displaystyle\sum_{N=1}^{N_T} M_{i,j}^{(N)}, & M_{i,j}^{(1)} \times \cdots M_{i,j}^{(N_T)} \neq 0 \end{cases} \end{aligned} \tag{9}$$

where $M^{(N_T)}$ represents the background weight maps of the N_T layers. To facilitate computation, FSAB1 in each subnetwork is selected to extract the corresponding background weight maps M, yielding a total of N_T background weight maps. Equation (9) demonstrates that the element (i, j) in M is 0 if any weight from $M^{(1)}$ to $M^{(N_T)}$ is 0. Conversely, if all the corresponding weights are nonzero, the mean value of the weights from $M^{(1)}$ to $M^{(N_T)}$ is calculated to obtain the final confidence background weight.

Put simply, the more unlikely it is for a pixel to be a part of the background, the more probable it is for it to be an anomaly. Consequently, if a pixel is not included in the background weight map M in any of the FSABs, it should be assigned a high weight to receive greater attention during AD. This process can be represented as follows:

$$\begin{aligned} \overline{M_{i,j}} &= 1 - M_{i,j}^{(0)} \cup M_{i,j}^{(1)} \cdots \cup M_{i,j}^{(N_T)} \\ &= 1 - \frac{1}{N_T}\sum_{N=1}^{N_T} M_{i,j}^{(N)} \end{aligned} \tag{10}$$

where $\overline{M_{i,j}}$ is referred to as the conjugate background weight. A higher value of $\overline{M_{i,j}}$ implies a lower likelihood of the corresponding position being part of the background.

In addition, the attention distribution of anomalies tends to be more concentrated due to their sparsity, resulting in stronger local attention energy. Therefore, the local mean of the attention score can be utilized to quantify the local energy of the spatial attention distribution. This calculation can be expressed as

$$E_{i,j} = \frac{1}{N_W} \sum_{z\in[i\pm w, j\pm w]} A_{i,j2\ z}^{\text{spa_}t} \tag{11}$$

where $z \in [i \pm w, j \pm w]$ represents the spatial local neighborhood of position (i, j), which contains a total of N_W pixel points, and $E_{i,j}$ is the local attention energy map. If a pixel falls within the foreground region and exhibits a high level of local attention energy, it may be an anomaly. This procedure can be formulated as

$$F_{i,j} = E_{i,j} \times \overline{M_{i,j}} \tag{12}$$

where $F_{i,j}$ is referred to as the confidence foreground weight. During the testing stage, a higher value of $F_{i,j}$ indicates a higher probability of an anomaly at the corresponding location. Examples of confidence background weight maps and confidence foreground weight maps are presented in Table I.

TABLE I
FSA DETAILS

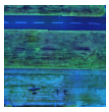
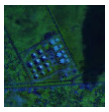
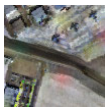
Image cube	Background weight map	Confidence foreground weight
		

C. Spectral Attention Mechanism

The low-rank nature of background pixels typically results in a stable interdependence among the spectral channels of the background. This interdependence can be effectively captured by employing a spectralwise self-attention layer, which enables the network to effectively restore the spectral content of the background. To this end, we employ the well-designed spectral self-attention technique [54].

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TABLE II
DATASET DETAILS

Dataset Details	Datasets					
	 Gulfport	 Pavia	 Texas Coast-2	 Los Angles-1	 San Diego-1	 San Diego-2
Resolution	3.4m	1.3m	17.2m	7.1m	3.5m	3.5m
Spatial Size	100×100	150×150	100×100	100×100	100×100	100×100
Spectral Overage Channels	400-2500nm	430-860nm	450-1350nm	430-860nm	370-2510nm	370-2510nm
Bands	191	102	207	205	189	189
Anomaly Material	Three Lanes	Vehicles	Buildings	Buildings	Three Lanes	Three Lanes
Anomaly Proportion	0.60%	0.28%	1.55%	2.72%	0.57%	1.34%

Specifically, the output features of the FSAB $X_{\text{out}}^{\text{spa}} \in \mathbb{R}^{k\times k\times n_s}$ are reshaped according to the spectral dimension, generating spectralwise tokens $X^r \in \mathbb{R}^{m\times n_s}$. The corresponding spectral dimensional query matrix Q^r, key matrix K^r, and value matrix V^r are then obtained through linear transformation. Similar to the FSAB module, the final aggregated features $X_{\text{out}}^r \in \mathbb{R}^{k\times k\times n_s}$ are achieved through normalization and residual connection. The procedure can be expressed as follows:

$$\begin{cases} Q^r = X^r W_Q,\ K^r = X^r W_K,\ \ V^r = X^r\ \ W_V \\ \widehat{X^r} = LN\left(\text{MSA}\left(Q^r, K^r, V^r\right)\right) + X^r \\ X_{\text{out}}^r = LN\left(\text{FFN}\left(\widehat{X^r}\right)\right) + \widehat{X^r} \end{cases} \tag{13}$$

where $\text{MSA}(Q^r, K^r, V^r)$ represents the calculation of spectral dimensional multiheaded self-attention.

The spectralwise self-attention mentioned earlier enhances the fusion of information among different spectral channels, resulting in more global output features X_{out}^r out. Furthermore, the aggregated spectral features are more likely to represent the background content, which enhances the discrimination ability of the model.

D. Training Loss and Anomaly Detection

The objective of the proposed model is to recover the background spectrum with high quality. Therefore, the quality of the recovered spectrum in the confidence background region is defined as the training loss, and the root-mean-square error (RMSE) is used to measure the quality difference between the generated HSI and the real HSI. The overall loss function can be expressed as follows:

$$\mathcal{L} = \frac{1}{\sqrt{N_p}} \| M \odot \left(S_i' - \hat{S}_i\right) \|_F \tag{14}$$

where $\odot$ denotes the dot product over spatial location, and $N_p = k \times k$ represents the number of pixels in the cube. The loss function [see (14)] comprises two components: the first component M represents the confidence background weight map, calculated using (9) and the second component is the standard RMSE.

To minimize computational overhead, during the testing stage, the input image is partitioned into nonoverlapping spatial cubes $X_{ti} \in \mathbb{R}^{k\times k\times 3}$, ensuring that they maintain the same size as those used in the training stage. Here, ti represents the cube index in the test stage. Anomalous targets are detected by leveraging the self-supervised difference recovery property and the spatial distribution characteristics. A pixel with a high confidence foreground weight and a significant recovery error is more likely to be an anomaly. To calculate the anomaly score, a weighted sum of these two terms is used

$$\text{Ano}\left(S_{i,j}'\right) = \| S_{i,j}' - \widehat{S_{i,j}} \|_2 + \alpha \cdot F_{i,j} \tag{15}$$

where α is the weighting coefficient to balance their relationship. Finally, similar to prior research [25], [28], [55], we apply normalization to the detection map within the interval of [0, 1]. Following the smoothing process of the detection map, we apply the sigmoid function to serve as the normalization function. Building upon the preceding context, the conclusive summary of the proposed method is outlined in Algorithm 1.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Datasets

We use six public datasets, namely, Gulfport, Pavia, Texas Coast-2, Los Angeles-1,[1] San Diego-1, and San Diego-2, to evaluate the proposed method. These datasets contain various types of anomalies with different attributes, and their details are listed in Table II.

B. Experimental Setup

1) Comparison Methods: To verify the effectiveness and superiority of our proposed method, we compare it with eight existing methods, including RX [1], LRX [3], CRD [11], fractional Fourier entropy (FrFE) [12], attribute and edge-preserving filtering-based detection (AED) [19], HADGAN [30], Wasserstein generative adversarial networks (WGAN) [23], and background-anomaly separability with a generative adversarial network (BASGAN) [26], which represent classical and state-of-the-art techniques in the field of HAD. RX, LRX, and AED are representative of

[1] http://xudongkang.weebly.com/

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Algorithm 1 SSHADFormer for HSI AD

Input: Given a hyperspectral image S;

Initialization:

1: Generating the corresponding RGB image X;

2: Performing sliding window operation to generate X_i and S_i by Eq.(1);

Training Model:

While not converged do

1: Network forward, get feature map X_i^0;

For each Finite Spatialwise Attention Block:

2: Conv/Deconv layer;

3: Obtain cluster centers according to Eq. (7);

4: Compute Finite Spatialwise Attention by Eq. (4);

5: Obtain Spectral Attentive layer (SA) by Eq. (13);

6: Feed Forward (FF);

Loss function:

7: Calculate the loss function according to Eq. (14);

End

Testing:

1: Partition the image into cubes $X_t i$ and $S_t i$;

2: Calculate the anomaly score by Eq. (15).

statistics-based approaches, while CRD and FrFE are classical representation-based methods. HADGAN, WGAN, and BASGAN are among the leading-edge deep learning-based techniques. The comparative analysis results are presented in Table III.

2) Evaluation Criterion: We utilize commonly used evaluation metrics, namely, the 3-D-receiver operating characteristic (3D-ROC) curve [56] and the area under curve (AUC) [57], to assess the effectiveness and superiority of the proposed method.

3) Implementation Details: The proposed method is implemented using Python 3.8 and based on the PyTorch deep learning framework (version 1.8). The experiments are conducted on a workstation with an Intel Core i7-10700 CPU, an NVIDIA 2080Ti GPU, and 32 GB of RAM. A batch size of 4 is used during the training process. For consistency, we utilized the hyperspectral imaging library in the MATLAB 2022 Image Processing Toolbox as a tool to generate color images for each dataset. This was achieved by inputting the spectral information parameters of each dataset.

Specifically, the process of generating RGB cubes involves three distinct stages: data loading, parameter setting, and culminating in the creation of RGB cubes.

1) *Data Loading:* Load an HSI and the corresponding spectral wavelength information for each spectrum and generate corresponding hypercube objects.
2) *Parameter Setting:* Open the hyperspectral viewer function in the hyperspectral imaging library, import the corresponding hypercube objects and choose RGB as the default bands. This tool can provide recommended band numbers corresponding to R, G, and B channels. The example process is illustrated in Fig. 7.
3) *Creation of RGB Cubes:* Based on the band numbers provided in the preceding steps, choose the corresponding content of the HSI in MATLAB and proceed with programming to generate RGB cubes.

C. Detection Performance

Fig. 4 illustrates the graphical detection results of several approaches on each dataset. The 3-D and 2-D ROC curves of nine methods on six datasets are presented in Fig. 5. In order to evaluate the separability of our proposed method compared with several other methods, we plotted the background-anomaly separation maps in Fig. 6. To quantitatively analyze the performance of the proposed approach, the AUC values of each method are provided in Tables IV and V, with the top-performing indicators highlighted in bold.

Regarding the Gulfport dataset, Fig. 4(a) displays the detection results. The homogenous spectrum of the anomalous target, which is the white lane marking at the top of the image, poses a challenge to AD. The LRX approach fails to preserve the form of the target and misses some anomalies, as shown in Fig. 4. Although the CRD, AED, and HADGAN methods yield more distinct detection results, their background suppression is weak, and the detection result map contains more background information. The RX, FrFE, and BASGAN methods help avoid false positives, but certain anomalies are still missed. The WGAN method detects anomalous targets with low response energy and insufficiently obvious abnormalities. In contrast, the proposed method precisely detects all anomalies and better maintains the target's structure, even though the Gulfport dataset has a few anomalies. The ROC curves in Fig. 5(a1) reveal that the proposed method achieves higher detection accuracy than the other methods. Moreover, the proposed approach has a low false positive rate, as demonstrated by the ROC curves. The results in Fig. 6(a) clearly demonstrate that the proposed method outperforms the other methods in terms of background-anomaly separation. Tables IV and V provide independent detection rates and false detection rates of each method on the Gulfport dataset. The proposed approach yields the highest score of 99.18%, as shown in Tables IV and V. Additionally, it exhibits lower false detection rates than most of the baseline methods.

Regarding the Pavia dataset, the detection results are shown in Fig. 4(b). Due to the narrow target region and sparse distribution of anomalies, AD is challenging for this dataset. The LRX, AED, CRD, and FrFE methods are significantly disturbed by the background, while the object outlines in the detection maps are clearly visible. The RX, HADGAN, and WGAN methods can identify anomalies, but the detection results are less visible subjectively due to low response strength. The BASGAN method eliminates almost all visual anomalies. Intuitively, the proposed method outperforms many others in terms of both background suppression and the ability to detect all anomalies. The proposed method also achieves effective background-anomaly separation for the Pavia dataset when compared with other algorithms, as shown in Fig. 6(b). Table IV provides the AUC values for the eight baseline approaches and the proposed method. Table IV shows that the proposed method, WGAN, and AED are the top three methods, with respective rankings of 98.27%, 97.96%, and 97.92%. The first-ranked method improves by 0.31% over the second-ranked method, and the second-ranked method improves by 0.04% over the third-ranked method. In other words, the improvement of the proposed method

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TABLE III
COMPARATIVE ANALYSIS OF BENCHMARKS

Category	Sub-category	Representative Method	Feature			Objective	
			Learned feature	Architecture	Basic computing unit	Criterion	Paradigm
Background model-based methods	RX	RX	No	–	Image	Deviation from background distribution	Unsupervised
	Improve-RX	LRX	No	–	Cube	Deviation from local distribution	Unsupervised
Background model-based methods	Feature transformation	AED	No	–	Image	Small areas/ Distinct spectral signatures/ Local consistency	Unsupervised
	Collaborative representation	CRD	No	–	Image	Spatial neighborhood representation	Unsupervised
	Entropy-based representation	FrFE	No	–	Image	Sensitivity to FT changes	Unsupervised
Deep learning-based methods	Discriminative reconstruction	HADGAN	Yes	AE+ discriminator	Pixel	Small areas/ distinct spectral signatures	Unsupervised
	Semi-supervised GAN	WGAN	Yes	AE+ discriminator	Pixel	Background distribution estimation	Semi-supervised
	Pseudo-label	BASGAN	Yes	AE+ 2discriminator	Pixel	Saliency map extraction	Semi-supervised
	Proposed method		Yes	Transformer	Cube	Spectral reconstruction/ learning-based background model	Self-supervised

over the second-ranked method is 7.75× more significant than the improvement of the second-ranked method over the third-ranked method.

Regarding the Texas Coast-2 dataset, the detection results are shown in Fig. 4(c). The RX, AED, FrFE, HADGAN, and WGAN methods produce similar subjective effects, all being significantly impacted by the background. While some anomalies are missed by these methods, the LRX, CRD, and BASGAN approaches show excellent background suppression and fewer false detections. In contrast, the proposed method can accurately identify anomalies and provide a more detailed description of their geometry. Fig. 6(c) illustrates that the anomalies detected by the proposed method can be more easily separated from the background than those detected by other methods. The proposed approach ranks third, as demonstrated in Tables IV and V. The top two methods, FrFE and HADGAN, have poor visual effects but a marginal numerical advantage. However, while the proposed method performs similarly to the LRX and WGAN methods in terms of overall accuracy, it exhibits lower false detection rates. Furthermore, the new method outperforms the other two methods in terms of AD rate.

Regarding the Los Angles-1 dataset, the detection results are presented in Fig. 4(d). While certain background areas are mistakenly identified as anomalies by RX, FrFE, HADGAN, and WGAN, LRX, CRD, and BASGAN demonstrate partial anomaly misses that leave portions of the target's form unpreserved. The AED algorithm performs a better job of highlighting abnormalities but fails to precisely define their shape. Meanwhile, compared with the other detection methods, the proposed method also achieves effective background-anomaly separation, as depicted in Fig. 6(d). In comparison with other methods, the proposed approach offers the most accurate and notable detection results. Numerical analysis in Table IV reveals that the proposed method improves detection rates by an average of 9.60% compared with other approaches. Although the LRX, CRD, and BASGAN methods obtain a low $AUC_{(\mathrm{FPR},\tau)}$ for false detection performance evaluation, they have significant leakage issues that compromise detection accuracy. Overall, it appears that the proposed method outperforms existing benchmark methods in terms of detection performance.

Regarding the San Diego-1 dataset, Fig. 4(e) presents the detection results. RX and LRX methods exhibit poor performance with anomalous omissions. AED, CRD, HADGAN, and WGAN methods can detect anomalies effectively, but cannot efficiently suppress background interference. The FrFE and BASGAN approaches achieve better background suppression but cannot maintain information about the target edge shape. In contrast, the proposed method can precisely locate each aircraft (anomalous target), recognize the wing and fuselage structure, and identify a highly distinct contour of the aircraft. As a result, our method demonstrates superior background-anomaly separation capabilities. As seen in Fig. 5(e2), the curve of the proposed method remains consistently above other approaches. The proposed method identifies San Diego-1 with a detection accuracy of 99.62%, which is the highest of all methods according to Table IV. The detection accuracy is 11.29% higher than the lowest-ranked LRX method and 0.21% higher than the second-ranked AED. Table V shows that the proposed method surpasses the most sophisticated methods for background interference suppression. Overall, the proposed method demonstrates the optimum detection performance in both detection outcomes and performance assessment outcomes.

Regarding the San Diego-2 dataset, the detection results are presented in Fig. 4(f). Accurately recognizing the targets in

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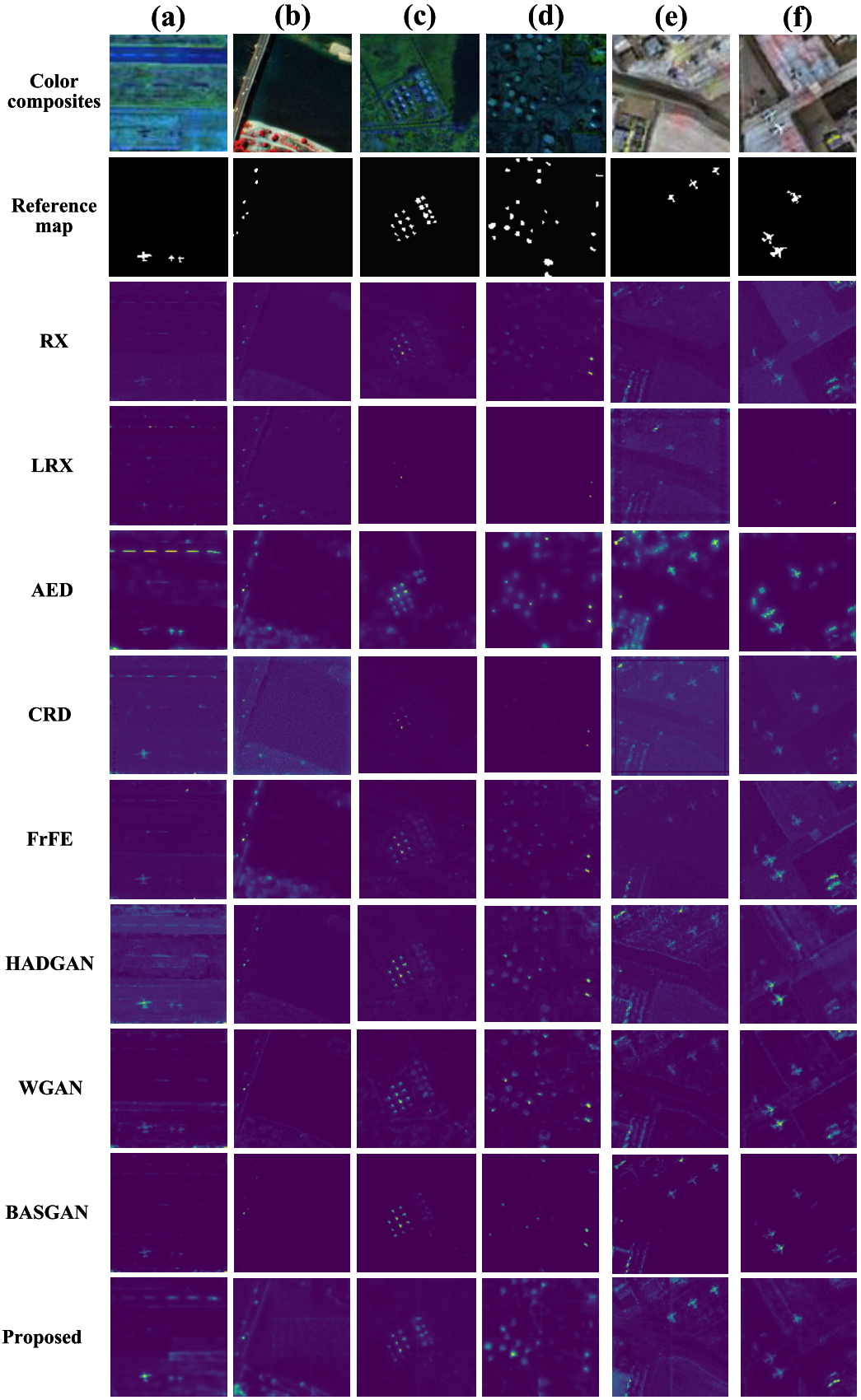


Fig. 4. Color composites map, reference map, and detection maps of the compared methods for six different datasets. (a) Gulfpor. (b) Pavia. (c) Texas Coast-2. (d) Los Angeles-1. (e) San Diego-1. (f) San Diego-2.

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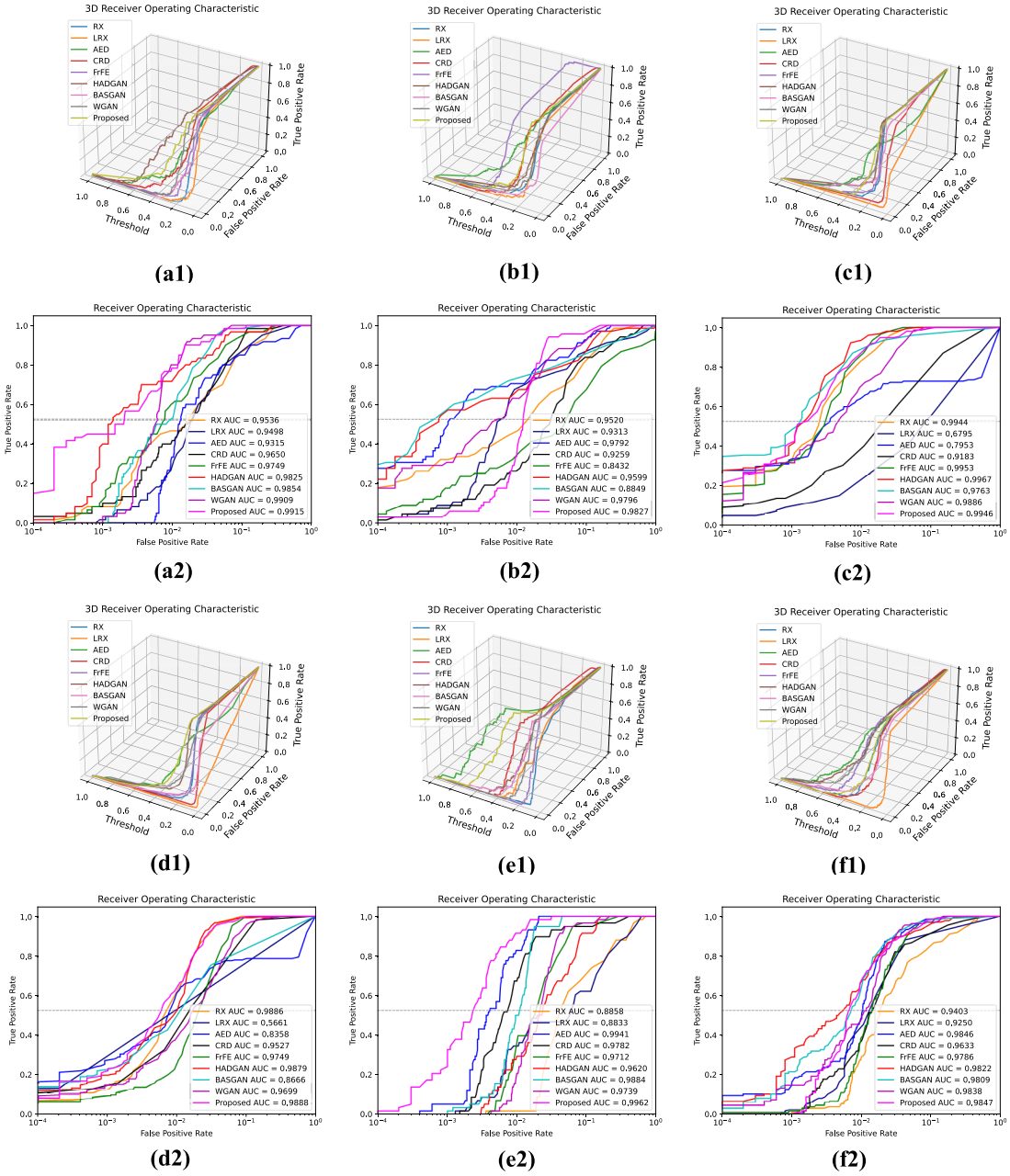


Fig. 5. ROC curves for the nine methods on (a) Gulfport, (b) Pavia, (c) Texas Coast-2, (d) Los Angeles-1, (e) San Diego-1, and (f) San Diego-2. From left to right are: (a1)–(f1) 3-D ROC curves and (a2)–(f2) 2-D ROC curves of (TPR and FPR).

this dataset is challenging because some background features, such as hangars, have similar geomaterials to the anomalous targets, i.e., airplanes. RX, AED, CRD, FrFE, HADGAN, and WGAN fail to completely isolate the anomalies from the background. LRX and BASGAN achieve better background suppression but at the cost of losing some anomalies. In contrast, the proposed method not only accurately locates the three airplanes (anomalous targets), but also precisely delineates their entire forms. Compared with other methods, the proposed method produces the most visually similar output to the reference map. It can be said that our method has good background-anomaly separation. The ROC curve of the proposed method almost completely encloses the ROC curves of other methods. The proposed method achieves the highest detection accuracy of 99.85% on the San Diego-2 dataset, outperforming previous benchmark methods by an average of 1.74%. According to Table V, compared with the LRX method, which has the best background suppression on this dataset, the proposed method improves detection accuracy by 5.97% while also achieving competitive performance in suppressing background interference. Overall, the results demonstrate that the proposed method is a highly competitive AD approach. According to Table V, the proposed method achieves a 5.97% improvement in detection accuracy compared with the LRX method, which has the best background suppression performance for the San Diego-2 dataset.

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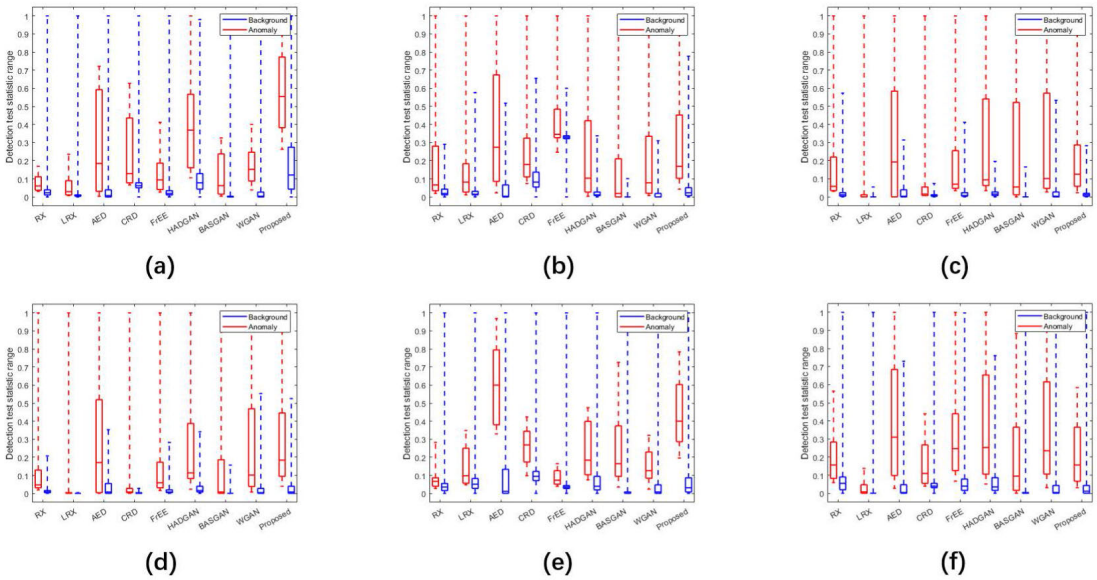


Fig. 6. Separability maps for the nine methods on (a) Gulfport, (b) Pavia, (c) Texas Coast-2, (d) Los Angeles-1, (e) San Diego-1, (f) San Diego-2.

TABLE IV
AUC SCORES OF TPR AND FPR

Methods	Datasets						Average
	Gulfport	Pavia	Texas Coast-2	Los Angles-1	San Diego-1	San Diego-2	
RX	0.9536	0.9520	0.9944	0.9886	0.8858	0.9403	0.9525
LRX	0.9498	0.9313	0.6795	0.5661	0.8833	0.9250	0.8225
AED	0.9315	0.9792	0.7953	0.8358	0.9941	0.9846	0.9201
CRD	0.9650	0.9259	0.9183	0.9527	0.9782	0.9633	0.9506
FrFE	0.9749	0.8432	0.9953	0.9749	0.9712	0.9786	0.9564
HADGAN	0.9825	0.9599	**0.9967**	0.9879	0.9620	0.9822	0.9785
WGAN	0.9909	0.9796	0.9763	0.9699	0.9739	0.9838	0.9791
BASGAN	0.9854	0.8849	0.9886	0.8666	0.9884	0.9809	0.9491
Proposed	**0.9918**	**0.9827**	0.9946	**0.9888**	**0.9962**	**0.9847**	**0.9898**

Notably, the proposed method also demonstrates competitive performance in suppressing background interference. These findings strongly suggest that the proposed method is a highly competitive AD approach.

The performance evaluation on all six datasets indicates that the proposed method demonstrates good detection performance. In comparison with the eight baseline methods, the overall detection results of the proposed approach are superior, implying that it outperforms the mainstream HAD techniques to a certain extent.

D. Model Agility Analysis

The agility of a method for anomalous pixel detection refers to its level of dexterity. To verify the agility of the proposed method, we conducted testing experiments using synthetic hyperspectral datasets. These datasets were created by blending small, imperceptible anomalous spectral vectors into a real-hyperspectral dataset. The proposed method's agility was evaluated on these two synthetic datasets and compared with the deep learning methods HADGAN, WGAN, and BASGAN.

To construct synthetic datasets for evaluating the proposed method's performance, we followed the following steps. First, we used the San Diego-2 dataset as the basic data and extracted all anomalous pixels from the San Diego-1 and San Diego-2 datasets. We then averaged the extracted anomalous pixels to construct anomalies T_1 and T_2, respectively. Second, we selected the initial background pixels at the corresponding position as the background pixels B in San Diego-2. Third, we constructed the synthetic anomaly S by fusing the construct anomalies ($T = T_1 or T_2$) and background pixel B using the calculation in (16). Finally, we replaced the background pixels B at the corresponding positions in the San Diego-2 dataset with the synthetic anomaly S. We defined different synthetic datasets based on the blended anomalies, with the dataset containing the San Diego-1 anomaly T_1 named San Diego-2_synthetic1, and the dataset containing the San Diego-2 anomaly T_2 named San Diego-2_synthetics2

$$S = B \times (1 - \kappa) + T \times \kappa. \tag{16}$$

The experiment included four different groups with varying weights κ, where κ was set as 0.4, 0.6, 0.8, and 1. Synthetic

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TABLE V
AUC SCORES OF FPR AND τ

Methods	Datasets						Average
	Gulfport	Pavia	Texas Coast-2	Los Angles-1	San Diego-1	San Diego-2	
RX	0.0266	0.0252	0.0155	0.0133	0.0400	0.0606	0.0302
LRX	0.0105	0.0220	**0.0022**	**0.0020**	0.0581	**0.0027**	0.0163
AED	0.0240	0.0217	0.0160	0.0220	0.0478	0.0210	0.0254
CRD	0.0682	0.0929	0.0092	0.0026	0.0970	0.0459	0.0526
FrFE	0.0247	0.3303	0.0146	0.0132	0.0388	0.0507	0.0787
HADGAN	0.0884	0.0171	0.0186	0.0223	0.0541	0.0466	0.0412
WGAN	**0.0054**	0.0069	0.0025	0.0132	**0.0097**	0.0046	**0.0071**
BASGAN	0.0126	**0.0021**	0.0118	0.0022	0.0210	0.0208	0.0118
Proposed	0.0237	0.0331	0.0136	0.0194	0.0459	0.0219	0.0262

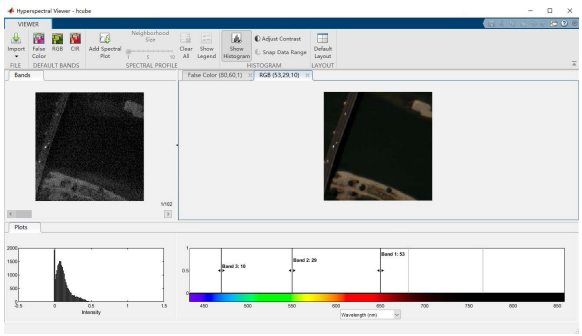


Fig. 7. Example of the hyperspectral viewer operation process.

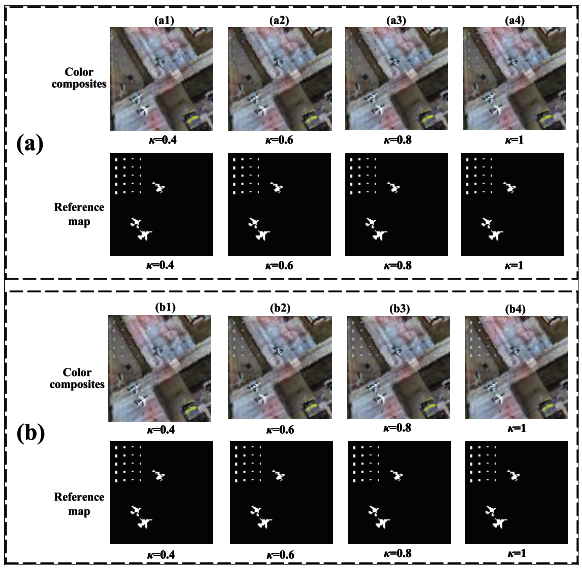


Fig. 8. Color composites of the HSIs and reference maps of (a) San Diego-2_synthetic1 and (b) San Diego-2_synthetic2.

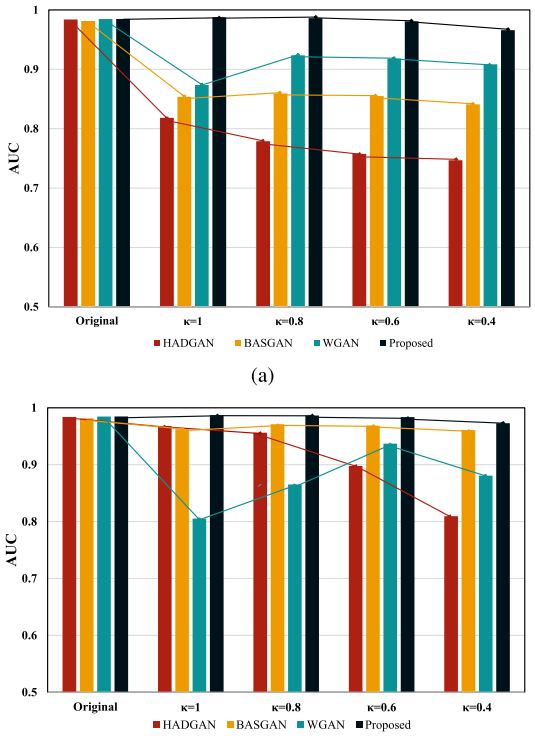


Fig. 9. Quantitative comparison of the detection results on (a) San Diego-2_synthetic1 and (b) San Diego-2_synthetic2 with different κ using different methods.

anomalies were inserted into the upper left background area of the image in the synthetic dataset and arranged in an array. A total of 20 synthetic anomalies were inserted, arranged in five rows and four columns, with sizes per column from left to right: 3×2, 2×2, 2×1, and 1×2 pixels, respectively. The San Diego-2_synthetic1 dataset and San Diego-2_synthetic2 dataset were constructed and are shown in Fig. 8(a) and (b), correspondingly. As seen in Fig. 8, the color composites and reference maps of the two synthetic datasets are given, where columns (a1)–(a4) and (b1)–(b4) represent different mixing weight coefficients κ, from left to right: $\kappa = 0.4$, 0.6, 0.8, and 1.

The detection accuracies of different methods on the San Diego-2_synthetic1 and San Diego-2_synthetic2 datasets with varying κ values are presented in separate plots in Fig. 9(a) and (b). Fig. 9 quantitatively compares the detection effects of different methods under different conditions, demonstrating the effectiveness and adaptability of the methods in detecting small anomalies. The fluctuation in detection performance of the four methods with respect to the value of κ is shown in Fig. 9. When κ is small, the target content tends to blend with the background, requiring the methods to be highly agile. For clarity, Table VI lists the average change

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TABLE VI
AMOUNTS OF RELATIVE CHANGES IN PERFORMANCES WITH DIFFERENT κ

		HADGAN	BASGAN	WGAN	Proposed
Average AUC Change	San Diego-2_$synthetic1$	0.2086	0.1291	0.0787	**0.0046**
	San Diego-2_$synthetic2$	0.0761	0.0156	0.1125	**0.0023**

TABLE VII
COMPREHENSIVE COMPARATIVE ANALYSIS

Sub-category	Representative method	Advantage	Disadvantage
Discriminative Reconstruction	HADGAN	High accuracy	Moderate agility
Semi-supervised GAN	WGAN	High accuracy, good stability	Moderate agility
Pseudo-label	BASGAN	Good stability	Moderate accuracy
Proposed method		High accuracyGood agility	Moderate stabilit

in AUC compared with the baseline detection performance of the four methods when the synthetic weight κ takes different values in the two datasets. The initial AUC of each method on the San Diego-2 dataset is used as the baseline performance. The minimum change fluctuation value is highlighted in bold in Table VI.

As illustrated in Fig. 9, the proposed method outperforms the other three deep learning methods in detecting small anomalies with different values of κ on both datasets. Moreover, the proposed method exhibits small numerical fluctuations as κ continuously changes, while the other three methods all exhibit large numerical fluctuations. Additionally, Table VI shows that the detection accuracy of the proposed method remains stable for different values of κ on the San Diego-2_synthetic1 dataset. Compared with the baseline detection accuracy, the proposed method has an average AUC change of 0.46%, while HADGAN, WGAN, and BASGAN have average AUC changes of 20.86%, 12.91%, and 7.87%, respectively. For the San Diego-2_synthetic2 dataset, the proposed method yields an AUC change of 0.23%, while HADGAN, WGAN, and BASGAN still have larger average AUC changes of 7.61%, 1.56%, and 11.25%, respectively. In summary, the proposed method demonstrates better performance in detecting small anomalies and exhibits good detection agility compared with the other methods.

Table VII provides a comprehensive comparison of the four deep learning methods. Results demonstrate that the proposed SSHADFormer method achieves superior accuracy and outperforms others in capturing anomalies. However, there is still room for improvement in terms of stability. Notably, the WGAN method stands out for its enhanced stability achieved through a saliency analysis process. Nonetheless, this comes at the expense of ability.

E. Ablation Study

In this section, ablation experiments are carried out on the San Diego-2 dataset to verify the reasonableness and effectiveness of parameter selection, which includes window size, hyperspectral dimension, clustering number, and weighting coefficients.

TABLE VIII
RELATIONSHIP BETWEEN WINDOW SIZE AND SYSTEM PERFORMANCE ON SAN DIEGO-2 DATASET

	Performance		
Window Size	10	20	50
AUC	0.9113	0.9572	0.9844

1) Window Size Selection Verification Experiment: The sliding window size has a significant impact on the results, and it is crucial to select an appropriate window size. To investigate the effect of window size on the performance of the proposed method, we conducted an experiment by setting three different window sizes: 10, 20, and 50 pixels, respectively. The results under different window sizes are summarized in Table VIII. It can be observed that the proposed method achieves the highest AUC of 98.44% when the window size is set to 50 pixels. We can further analyze that the larger window size helps the transformer network to capture more stable relationships in both spatial and spectral domains, thereby improving the AD performance of the model. However, it should be noted that a larger window size also incurs more computational complexity. Therefore, we set the window size to 50 pixels in the experiment.

2) Hyperspectral-Dimensional Validation Experiment: We investigate the impact of dimensionality reduction on spectral channels. To this end, we constructed HSI data with varying spectral dimensions by setting different output parameters of the OCF algorithm. The model was then trained using these data to obtain AD performance for different spectral sizes. The results are presented in Fig. 10, where the horizontal axis represents the spectral dimension and the vertical axis represents the AUC score that measures the method's performance. We observe that the spectral dimension has a limited effect on the method's performance, and the best performance is achieved when the spectral dimension is set to 31. Therefore, for the sake of fairness in the entire experiment, we set the spectral dimension to 31.

3) Clustering Number Validation Experiment: We investigate the impact of the clustering number on the performance

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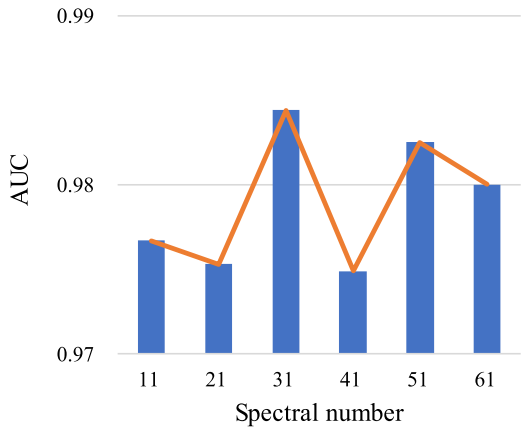


Fig. 10. Relationship histogram between spectral dimension and system performance on San Diego-2 dataset.

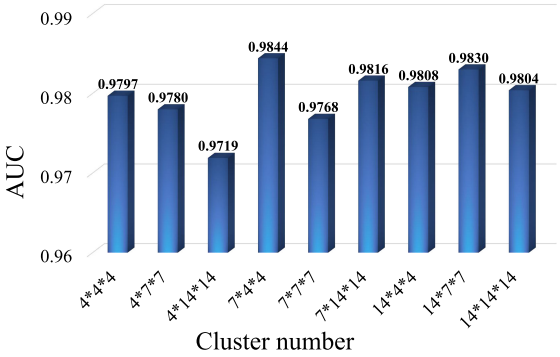


Fig. 11. Relationship histogram between clustering number and system performance on San Diego-2 dataset.

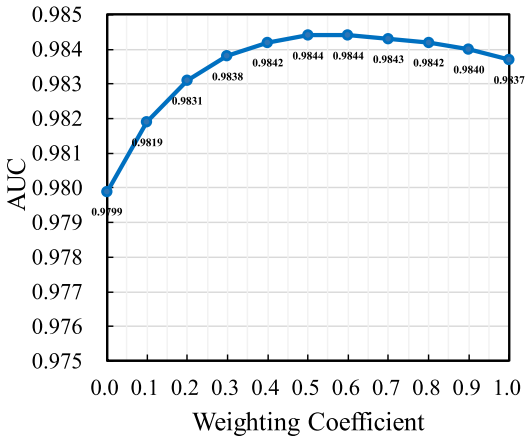


Fig. 12. Relationship between weighting coefficient and AD performance on the San Diego-2 dataset.

of our method. Specifically, we set a different number of clustering centers for each layer of the FSAB and obtain the corresponding AD performances. In this experiment, we assigned the same clustering number C_1 to FSAB1 and FSAB5 and the same clustering number C_2 to FSAB2 and FSAB4, while FSAB3 has a different clustering number C_3. We adjusted the combination of C_1, C_2, and C_3 in an iterative manner with discrete values of 4, 7, and 14, respectively, i.e., $C_1, C_2, C_3 \in [4, 7, 14]$. We set the intermediate layers to the same clustering number ($C_2 = C_3$), resulting in a total of nine different combinations. The results are presented in Fig. 11, where the horizontal axis indicates the clustering number of different layers and the vertical axis represents the performance of our method with different clustering numbers. Based on Fig. 11, our method achieves the best performance when the clustering numbers C_1, C_2, and C_3 are set to 7, 4, and 4, respectively. Therefore, the clustering numbers in the network are set to 7, 4, and 4, respectively, throughout the experiment.

4) Weighting Coefficient Validation Experiment: We evaluate the impact of the weighting coefficient α in (15) and validate the efficacy of the proposed confidence foreground weight. Specifically, we evaluate the corresponding performance by setting various weighting coefficients in this experiment and present the results in Fig. 12. The horizontal axis in Fig. 12 represents the value of the weighting coefficient, which takes values within the range [0, 1] with an interval of 0.1, and the vertical axis indicates the corresponding AUC scores. It can be observed that the AUC score tends to increase and then decrease with an increase in the weighting coefficients, and the optimal performance is achieved when the weighting coefficient is set to 0.5. Therefore, to ensure fairness throughout the entire experiment, the weighting coefficient is set to 0.5. In addition, when the weighting coefficient is 0, the confidence foreground weight is not applied. Comparatively, we find that the confidence foreground weight is effective and improves AD performance.

V. Conclusion

This article presents a self-supervised HAD algorithm based on FSA. The main findings and limitations of our study are summarized as follows.

1) Our proposed method utilizes a self-supervised mapping process from the visible spectrum to the hyperspectral spectrum to acquire and utilize high-level semantic information.
2) SSHADFormer extracts both spatial and spectral information from cube-level units, enabling an end-to-end AD approach. The FSA mechanism is introduced to eliminate anomalous interference during training and improve discrimination between background and anomalies from a spatial distribution perspective.
3) Extensive experiments demonstrate the superior performance and agility of the proposed method compared with eight state-of-the-art methods.
4) However, a notable limitation of our method is its inability to achieve low false alarm rates across various datasets. Our analysis indicates that the lack of cross-scenario stability inherent in a single self-supervised method is responsible for suboptimal performance. Going forward, we aim to address this limitation by exploring the incorporation of multiple self-supervised tasks. This strategy holds the potential to enhance our method's overall performance and achieve more favorable results.
5) In addition to using RGB images as inputs, the content and dimensions of the input images are subject

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to exploration, such as the inclusion of near-infrared bands. In addition, further investigation is needed for adaptive spectral band selection methods applicable to self-supervised HAD.

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